

Annexure-I

ACADEMIC TASK 1 SYNOPSIS

On

AI-BASED CLASS MONITOR SYSTEM

Submitted by

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**MASTER OF COMPUTER APPLICATIONS
(MCA)**

Course Code and Title: CAP679 - NEXT GEN COMPUTING PROJECT 2

Under the Supervision of

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**SCHOOL OF COMPUTER APPLICATIONS
LOVELY PROFESSIONAL UNIVERSITY
PUNJAB**

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1. TASK OVERVIEW AND SUBMISSION INSTRUCTIONS

Academic Task	Problem Identification and Gap Analysis
Objective	To identify and validate a significant real-world problem through literature review, stakeholder analysis and evidence collection.
Nature / Mode	Individual / Offline
Maximum Marks	30
Allotment / Submission	Week 2 / Week 5
Bloom's Taxonomy Level	Analyze (L4)

General Instructions

1. Use this prescribed sequence and retain all headings.
2. Select a specific, relevant and practically significant real-world problem.
3. Support claims with credible evidence such as scholarly studies, official reports, surveys, interviews or observations.
4. Review recent and relevant literature and compare existing solutions before establishing the gap.
5. Use a consistent citation style (IEEE or APA, as directed by the faculty mentor).
6. Recommended synopsis length: 8-12 pages, excluding cover page, annexures and plagiarism/similarity report.
7. Similarity and use of generative AI must follow institutional policy. Students remain responsible for accuracy and originality.
8. Obtain the faculty mentor's approval before proceeding to Academic Task 2.

Formatting Guidelines

Item	Requirement
Page size	A4
Font	Times New Roman, 12 pt (suggested)
Line spacing	1.5
Margins	1 inch / 2.54 cm
Alignment	Justified body text
Page numbering	Bottom centre or bottom right

2. PROJECT TITLE

AI-BASED CLASS MONITOR SYSTEM

Repository: <https://github.com/khushicharan2005/Class-Monitary-System>

3. EXECUTIVE SUMMARY

This project began with a common classroom problem: one teacher cannot watch every student while also explaining a lesson, answering questions, or managing an examination. The proposed Class Monitor System uses a camera feed to flag frames in which a mobile phone appears near a student. It can work with live video or a saved recording. YOLOv5 locates people and phones, OpenCV handles the frames, and the Flask service passes useful results to a MySQL database and a PHP dashboard. Face matching is kept as an optional part of the project and will only be tested with consent. The system does not decide that a student has broken a rule. It simply saves a time-stamped event that a teacher or invigilator can inspect. This keeps judgement with the responsible person and also makes false detections easier to correct. Research on smartphone distraction shows why the problem deserves attention, while work on real-time object detection shows that a small prototype is practical. Manual supervision gives context but misses brief events; ordinary CCTV records everything but leaves someone to search long footage later. This project connects detection, event storage, and review in one limited workflow. The prototype will be tested indoors using staged classroom scenes. The final work will include the detection pipeline, saved event records, dashboard screens, basic reports, and results for accuracy, response time, privacy, and day-to-day usefulness.

4. BACKGROUND AND CONTEXT

Phones are part of everyday student life. They are useful for notes, communication, and learning resources, but the same device can draw attention away from a lecture or provide unauthorised help in an examination. The difficulty is especially clear in a large room. A teacher may be writing on the board, helping one student, or checking attendance when another student briefly looks at a phone. An invigilator faces a similar problem while handling answer sheets and other duties. A normal CCTV camera keeps a recording, but it does not point out the few moments that may need attention. This project explores whether a small computer-vision system can do that first level of sorting. It will locate people and phones, save the time and camera details, and show the event on a web page. Teachers and invigilators remain responsible for deciding what actually happened. Students also have a direct interest because recording, identification, access, and retention must be handled fairly and openly.

4.1 METHODOLOGY

Development will proceed in small, testable parts. camera_node.py reads frames from a webcam, IP camera, or video file and gives each frame a consistent size and time stamp. yolov5_module.py loads the YOLOv5 model, applies the chosen confidence settings, and returns boxes for people and mobile phones. face_module.py is separate from the main detector. It will run only during approved, consented tests; otherwise events stay anonymous. app.py connects these parts through Flask and serves event data. schema.sql creates the MySQL tables for camera sources, detections, users, and review status. analysis.py prepares counts and simple time-based summaries. The PHP dashboard then shows live status, saved events, and review controls.

Testing will use staged classroom clips recorded with permission. The clips will include normal movement, permitted phone handling, partly hidden phones, different lighting, and an empty room. Phone detection will be measured with precision, recall, and F1-score. I will also record processing delay, repeated alerts, and dashboard loading time. False alerts will be grouped by the conditions that caused them instead of being hidden inside one overall score. Only the minimum personal data needed for the test will be stored, with restricted access and a fixed deletion period.

4.2 PICTORIAL REPRESENTATION

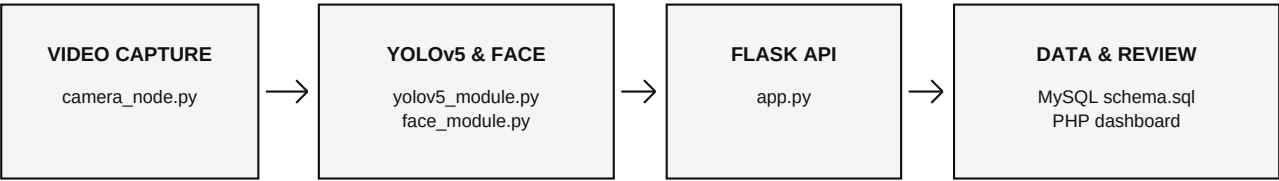


Figure 1. The camera module prepares each video frame. YOLOv5 detects people and phones, while face matching remains optional. Flask passes confirmed event data to MySQL, and the PHP dashboard presents it for review. analysis.py produces summaries from reviewed records.

Stage	Input	Principal processing	Output
Capture	Webcam/IP camera/video	Frame sampling, resize, timestamp	Normalised frames
Inference	Frames	Person/phone detection; optional face matching	Boxes, labels, confidence
Application	Detection result	Validation, event association, API logic	Structured event
Persistence & review	Event payload	MySQL storage; PHP dashboard; analysis.py	Review status and reports

5. PROBLEM IDENTIFICATION AND DEFINITION

5.1 Problem Statement

In a lecture or examination, supervision is only one of several jobs being done at the same time. A teacher may be explaining a topic, answering a question, or checking student work. An invigilator may be distributing papers or responding to a query. A phone can therefore appear for a few seconds without being noticed, particularly in a crowded room. Direct observation is valuable because a person understands context, but it is not continuous. CCTV provides a continuous record, yet finding one short event later can take considerable time. Missed or delayed review leads to uneven enforcement, distraction during class, and a possible route to unfair help during an examination. The project addresses this practical gap by marking frames that contain both a person and a phone, saving the time of the event, and giving the relevant clip or image to an authorised reviewer.

5.2 Need and Significance

The value of the project is practical as well as academic. It applies object detection, image processing, databases, and web development to a problem that can be observed in a real classroom. If the prototype works reliably, a reviewer will not need to watch an entire recording to locate a few possible events. More consistent review may also support classroom focus and examination fairness. These benefits depend on careful limits. A phone may be allowed for a class activity or may simply be lying on a desk. For that reason, every result must remain an alert rather than an accusation. Users should know when the system is active, who can see the records, and when those records will be deleted.

5.3 Supporting Evidence

Evidence Source	Key Observation / Finding	How It Validates the Problem
Sung, Chang and Liu [1]	Mobile-device use can support learning, yet classroom implementation introduces distraction and management concerns.	Establishes that educational benefit does not remove the need to govern off-task use.
Amez and Baert [2]	The reviewed literature reports a generally negative association between smartphone use and academic performance, although causality is complex.	Shows that problematic use warrants measured institutional attention.
Redmon et al. [3]	YOLO frames object detection as a single regression problem and supports real-time inference.	Supports the feasibility of video-based phone detection.
Teacher observation	Teaching and supervision compete for the same finite attention.	Explains why a prioritised alert can supplement, rather than replace, observation.
Conventional CCTV	Footage is continuous but normally requires retrospective human inspection.	Identifies the need for automatic event indexing and review.

6. LITERATURE REVIEW

The review covered three connected areas: what studies say about smartphone use in education, how real-time object detectors work, and what safeguards are needed when cameras or face data are involved. Older method papers were included where they explain the technology still used in this project. Recent review and governance sources were added to keep the educational and ethical discussion current. Throughout the review, a detected phone is treated as an object in an image, not as proof that a student was distracted or cheating.

Selection and Review Method

I searched scholarly databases and publisher sites using terms such as “smartphone classroom distraction,” “academic performance,” “YOLO object detection,” “face recognition privacy,” and “classroom computer vision.” Priority was given to peer-reviewed reviews, original detector papers, and recognised guidance such as the NIST framework. A source was kept when it helped explain the problem, choose an implementation method, plan an evaluation, or set a clear ethical limit. Unsupported product claims were not used as evidence.

Review Questions

- What does the literature establish about off-task smartphone use and learning conditions?
- Can a one-stage detector provide sufficiently prompt phone localisation for a classroom prototype?
- Which technical and contextual factors create false detections or missed events?
- What safeguards are required when video or biometric information may be processed?

Analytical Position

Taken together, the sources support a modest project claim. Off-task phone use can create a genuine classroom-management problem, and current detectors can locate phones in video quickly enough to test a prototype. They do not promise perfect performance. Camera angle, distance, obstruction, lighting, and the training data can all change the result. The project will therefore test an assisting tool in controlled conditions rather than present it as an autonomous surveillance system.

6. LITERATURE REVIEW — SUMMARY TABLE

Ref.	Author(s), Year	Method / Context	Key Findings	Limitation / Gap
[1]	Sung, Chang & Liu, 2016	Meta-analysis of mobile-device integration in teaching and learning	Mobile devices may improve learning outcomes when pedagogically integrated.	Does not isolate unauthorised phone use or classroom supervision.
[2]	Amez & Baert, 2020	Literature review on smartphone use and academic performance	Most reviewed studies report a negative association; mechanisms and causality require care.	Heterogeneous measures; observational evidence dominates.
[3]	Redmon et al., 2016	Unified, single-network object detector	YOLO enables rapid end-to-end object localisation and classification.	Original model trades some localisation accuracy for speed.
[4]	Ren et al., 2015	Region-proposal-based detector	Faster R-CNN achieves strong detection accuracy with learned proposals.	Two-stage design may be less suitable for modest real-time hardware.
[5]	Jocher et al., 2022	YOLOv5 implementation and training ecosystem	Offers practical pretrained models, configurable inference and deployment support.	Performance remains dependent on weights, classes and operating conditions.
[6]	Buolamwini & Gebru, 2018	Intersectional audit of commercial gender classification	Error rates differed substantially across demographic groups.	Task is classification rather than classroom face identification, but bias warning transfers.
[7]	NIST, 2023	AI Risk Management Framework	Advocates valid, reliable, transparent, privacy-enhanced and accountable AI.	High-level framework; controls must be adapted to the institution.
[8]	Bradski, 2000	OpenCV library overview	Establishes a practical foundation for real-time computer-vision development.	Library capability alone does not establish application accuracy.

6.1 SYNTHESIS OF LITERATURE

Studies of mobile devices in education do not support a simple “phones are good” or “phones are bad” conclusion. Sung, Chang, and Liu [1] found that mobile devices can improve learning when their use is planned as part of teaching. Outside that purpose, the same device can easily compete for attention. Amez and Baert [2] report that heavier smartphone use is usually associated with weaker academic performance, although the studies use different measures and cannot always establish cause. For this project, the reasonable conclusion is narrower: off-task phone use is a real management issue, but seeing a phone does not by itself prove harm or misconduct.

The computer-vision literature shows why real-time detection is worth testing. YOLO predicts object classes and bounding boxes in one pass [3], which makes it suitable when the result must appear quickly. Faster R-CNN [4] can give strong localisation, but its two-stage process is less convenient for the modest hardware planned here. YOLOv5 [5] is practical because pretrained weights and training tools are readily available. Its general “cell phone” class is not designed specifically for classrooms, so small, partly covered, or tilted phones may still be missed. OpenCV [8] provides the surrounding video capture and annotation tools.

A technical match is not the same as a fair judgement. A box around a phone cannot show whether its use was permitted, who owned it, or what the student intended. Adding face recognition makes the issue more sensitive. Buolamwini and Gebu [6] found unequal error rates across demographic groups in commercial classification systems, while NIST [7] stresses testing, transparency, privacy, and accountability. These findings support an optional face module, clear confidence values, limited access, short retention, and a reviewer who can reject an incorrect alert.

The literature supports a tool that helps a person search video; it does not support automatic punishment, hidden identity tracking, or indefinite storage. The useful contribution of this project is therefore the complete but limited workflow: capture a frame, detect a possible phone event, store it, and let an authorised user review it. Testing the connected workflow in a classroom setting matters more than reporting detector accuracy alone.

Synthesis conclusion

A fast detector is useful only when its mistakes and limits remain visible. Accuracy, response time, false alerts, privacy, and ease of review will therefore be assessed together.

7. ANALYSIS OF EXISTING SOLUTIONS

The comparison focuses on a simple question: how well does each approach help an authorised person notice and review a possible phone-use event in a classroom?

Solution / Approach	Core Features	Strengths	Limitations	Source
Direct teacher or invigilator observation	Immediate contextual judgement	No equipment; understands permission and intent	Attention is divided; events are not consistently recorded	Classroom practice
Conventional CCTV	Continuous live view and recording	Broad coverage; retrospective evidence	Requires human viewing; no automatic phone-event index	Operational baseline
Two-stage vision detector	Region proposals followed by classification	Potentially strong localisation accuracy	Greater compute and integration complexity for real-time use	Faster R-CNN [4]
Generic YOLO detector	Single-stage object detection	Fast inference; practical pretrained implementation	Small or occluded phones may be missed; domain shift	YOLO [3], YOLOv5 [5]
Proposed integrated system	Capture, YOLOv5, optional face module, Flask API, MySQL and PHP dashboard	Focused workflow; time-stamped review; modular design	Prototype depends on camera position, consent, weights and hardware	Current project

7.1 Comparative Interpretation

Direct observation remains best for understanding context, but a teacher or invigilator cannot look everywhere continuously. CCTV fills that time gap, although it creates hours of footage that still need to be watched. Object detectors can mark likely phone locations, but a detector by itself does not create a record that staff can review and resolve. The proposed system joins these steps. YOLOv5 was chosen because it can run promptly and is realistic for a student prototype, not because it is always more accurate than a two-stage detector. Classroom tests will show where it fails. The dashboard will display the image, time, and confidence so that a reviewer can dismiss a false alert rather than accept the model output automatically.

8. GAP ANALYSIS

Identified Gap	Evidence / Source	Impact of Gap	Opportunity for Project
Manual supervision is intermittent	Classroom practice; stakeholder analysis	Brief events may be missed while attention is elsewhere.	Generate prioritised, time-stamped alerts for review.
CCTV records but does not interpret	Operational comparison	Large footage volume delays identification of relevant events.	Index footage using phone/person detections.
Generic detectors are not classroom-calibrated	YOLO literature [3], [5]	Small, occluded or distant phones can produce misses and false alerts.	Evaluate thresholds and conditions on staged classroom video.
Detection is separated from review workflow	Existing-solution comparison	A bounding box alone does not support accountable follow-up.	Integrate Flask, MySQL, PHP dashboard and status tracking.
Identity processing creates privacy and bias risk	Buolamwini & Gebru [6]; NIST [7]	Uncritical deployment may cause unfair or intrusive monitoring.	Make face recognition optional, consent-based and access-controlled.
Behaviour cannot be inferred from object presence	Literature synthesis	A detected phone may be permitted or unrelated to misconduct.	Preserve human judgement and label outputs as possible events.

8.1 Final Gap Statement

No widely available classroom prototype found in this review combines the exact set of functions needed here: real-time person and phone detection, optional identification with consent, a structured event record, and a simple human review screen. Manual supervision understands context but cannot be continuous. CCTV records continuously but does not identify the moments worth checking. Research detectors locate objects, yet they usually stop before the practical questions of storage, access, correction, and review. This project will connect YOLOv5 and OpenCV to a Flask service, MySQL records, basic analysis, and a PHP dashboard. Its purpose is not to declare misconduct. It is to give an authorised teacher or invigilator a manageable list of possible events, together with enough evidence to accept or dismiss each one.

9. STAKEHOLDER AND USER ANALYSIS

Stakeholder / User	Role / Interest	Need / Pain Point	Evidence-Collection Method
Teachers	Primary reviewer during instruction	Low-disruption alerts; context; rapid dismissal of false events	Semi-structured interview and task walkthrough
Invigilators	Examination supervision	Reliable time-stamped cues without automatic accusation	Scenario-based interview and observation
Students	People represented in footage	Fairness, notice, privacy, correction and limited retention	Anonymous questionnaire and consent discussion
School administrators	Policy, access and accountability	Clear roles, audit trail, retention and incident procedure	Policy-document review and interview
IT / Lab staff	Installation and maintenance	Stable cameras, manageable compute, logs and recovery	Technical checklist and prototype test
Faculty mentor	Academic and technical oversight	Traceable evidence, feasible scope and ethical controls	Periodic review and repository inspection

9.1 Key Insights from Survey / Interview / Observation

Early discussion of the classroom workflow suggests that an alert is helpful only when it is short, clear, and easy to dismiss. A teacher needs the image and time of the event, not just a running count. Students need notice, limited storage, and a way for an incorrect result to be challenged. Administrators need clear access rules, while technical staff need stable camera positions and services that can be restarted without losing records. These points are working assumptions, not survey findings. A formal, anonymous stakeholder exercise is planned for Academic Task 2.

Participants will receive an information sheet, and no hidden recording will be used. Names will not be collected unless someone asks to be contacted again. Observation notes will focus on whether alerts interrupt work, whether the evidence is understandable, and how false events are handled. Results will be combined rather than reported person by person. The findings will be used to revise requirements R4, R7, R8, and R9.

10. PROJECT SCOPE, OBJECTIVES AND PRELIMINARY REQUIREMENTS

10.1 Aim

To build and evaluate a classroom video tool that flags possible student phone-use events and gives authorised staff clear evidence to review.

10.2 Specific Objectives

1. Develop a reusable capture pipeline that accepts a webcam, network camera or recorded video and preserves frame timestamps.
2. Configure YOLOv5 to detect persons and mobile phones, visualise confidence-labelled bounding boxes, and measure precision, recall and latency on staged classroom scenes.
3. Implement optional, consent-based face matching as a separable module that can be disabled without affecting anonymous event detection.
4. Integrate a Flask API, MySQL event schema, PHP review dashboard and analysis routine for recording, filtering, resolving and summarising events.
5. Evaluate usability, false-alert conditions, privacy controls and operational feasibility, and document recommendations for a limited institutional pilot.

10.3 Scope and Boundaries

In Scope	Out of Scope / Constraints
Single-camera or selected recorded-video input in a controlled indoor environment; person and mobile-phone detection; bounding-box visualisation; optional consent-based face matching; time-stamped event storage; authorised PHP dashboard; basic summaries; staged evaluation.	Automatic disciplinary decisions; covert surveillance; emotion or intent inference; audio recording; institution-wide multi-camera deployment; guaranteed identification under all conditions; cloud-scale analytics; unrestricted biometric retention; replacement of teachers or invigilators.

10.4 PRELIMINARY REQUIREMENTS

ID	Requirement Description	Type	Priority
R1	The system shall acquire frames from a configured webcam, IP stream or video file and retain source timestamps.	Functional	Must
R2	The system shall detect persons and mobile phones and return class, confidence and bounding-box coordinates.	Functional	Must
R3	The system shall associate a phone detection with a candidate person using a documented spatial rule and mark the result as a possible event.	Functional	Must
R4	The system shall store time, camera, confidence, evidence reference and review status in MySQL.	Functional	Must
R5	The Flask API shall validate inputs and provide authorised endpoints for health, recent events and review updates.	Functional	Must
R6	The PHP dashboard shall filter events, show evidence and permit an authorised reviewer to confirm, dismiss or annotate an event.	Functional	Must
R7	Face matching shall be disabled by default unless informed consent and institutional approval are documented.	Non-functional / privacy	Must
R8	The prototype should process frames with an observable end-to-end delay suitable for near-real-time review on the test hardware.	Non-functional / performance	Should
R9	The system shall protect accounts and event records through authentication, least-privilege access, validation, audit logging and a defined retention rule.	Non-functional / security	Must

Acceptance Basis

R1-R6 will be demonstrated through an end-to-end test from video input to reviewer action. R7 and R9 will be assessed by configuration and access-control inspection. R8 will be measured as capture-to-display latency on declared hardware; no universal performance claim will be made. Detection quality will be reported using precision, recall and F1-score, with representative false positives and false negatives retained in the evaluation record.

11. EXPECTED OUTCOMES AND INITIAL FEASIBILITY

11.1 Expected Outcomes / Impact

- A working prototype that connects camera input, YOLOv5 detection, optional face matching, Flask, MySQL, the PHP review screen, and analysis.py reports.
- A repeatable set of staged tests, with results for precision, recall, F1-score, processing delay, duplicate alerts, and failures under different conditions.
- A review screen that helps a teacher or invigilator find relevant footage without handing judgement to the software.
- Written rules for consent, access, retention, and privacy that can be checked by the faculty mentor before any wider trial.
- A tested base for later additions, without presenting this student prototype as a finished institutional system.

11.2 Preliminary Feasibility

Dimension	Preliminary Assessment
Technical	Feasible for a prototype using Python, OpenCV, YOLOv5, Flask, MySQL and PHP. Main uncertainties are small-object accuracy, occlusion and available compute.
Operational / User	Feasible if alerts are concise, evidence is reviewable and the system can be paused or disabled by an authorised user.
Economic / Resource	Open-source software limits licence cost; existing webcam/lab hardware may suffice. A GPU is beneficial but not assumed.
Ethical, Privacy and Security	Conditionally feasible with notice, consent where identity is processed, least-privilege access, limited retention and no automated sanctions.
Time / Schedule	Feasible within Task 2 if work proceeds module-by-module and face matching remains optional rather than blocking core detection.
Sustainability / Social Impact	Reuse of existing hardware and event-based storage reduces resource demand; benefits depend on proportional, transparent use.

11.3 RISKS AND ASSUMPTIONS

Risk / Assumption	Likelihood	Impact	Mitigation / Validation
Small, distant or occluded phones are missed	Medium	High	Test multiple distances and angles; tune confidence; document recall limits.
Books or similar rectangles trigger false alerts	Medium	Medium	Inspect confusion cases; require person proximity and consecutive-frame confirmation.
Low light or camera vibration degrades inference	Medium	Medium	Specify placement and lighting; stabilise capture; test adverse scenes.
Face recognition creates privacy or bias harm	Medium	High	Keep optional and off by default; require consent; test cautiously; permit anonymous operation.
Unauthorised access exposes event images	Low-Medium	High	Authentication, role-limited access, secure configuration, audit trail and retention deletion.
Hardware cannot sustain desired responsiveness	Medium	Medium	Benchmark smaller weights and frame sampling; report hardware and achieved latency.
A detection is treated as proof of misconduct	Medium	High	Use “possible event” language; require human confirmation; train reviewers.
Assumption: staged scenes represent initial use	—	Medium	Describe sample conditions and avoid generalising beyond the tested environment.
Assumption: camera and participants are authorised	—	High	Obtain written institutional permission and participant notice before capture.

Risk Position

The most serious risks are not merely technical. A missed phone reduces utility, but an overconfident or unauthorised interpretation can harm a student. The prototype therefore prioritises transparent uncertainty, controlled evidence access and human review alongside model performance. Deployment beyond staged academic evaluation is contingent on institutional approval and a documented data-governance process.

12. WORK PLAN FOR ACADEMIC TASK 2

Activity	Responsible Member(s)	Target Week / Date	Expected Deliverable
Confirm use cases, consent boundary and acceptance measures	Khushi Charan; Faculty Mentor	Week 1 / 28 Sep-4 Oct 2026	Approved requirements and evaluation protocol
Set up repository, environment and sample capture	Khushi Charan	Week 1	Reproducible setup and consented sample clips
Implement camera_node.py and frame tests	Khushi Charan	Week 2 / 5-11 Oct	Stable webcam/video ingestion
Implement yolov5_module.py; tune thresholds	Khushi Charan	Week 2-3	Person/phone detections with metrics
Implement optional face_module.py safeguards	Khushi Charan	Week 3	Disable-by-default identity module
Create schema.sql and MySQL access layer	Khushi Charan	Week 4 / 19-25 Oct	Tables, constraints and sample event records
Implement app.py Flask API	Khushi Charan	Week 4-5	Validated health, event and review endpoints
Build PHP dashboard and review workflow	Khushi Charan	Week 5-6	Event list, filters, evidence and decisions
Implement analysis.py summaries	Khushi Charan	Week 6	Frequency, confidence and status report
Integration, security and failure testing	Khushi Charan; mentor review	Week 7 / 9-15 Nov	End-to-end test record and issue log
User walkthrough and staged evaluation	Khushi Charan; consenting participants	Week 8	Usability notes and detection dataset
Analyse results, revise and document	Khushi Charan	Week 9	Metrics, limitations and revised prototype
Final mentor review and Task 2 submission	Khushi Charan; Faculty Mentor	Week 10 / 30 Nov-6 Dec	Approved report, demo and repository

13. REFERENCES (IEEE FORMAT)

- [1] Y.-T. Sung, K.-E. Chang, and T.-C. Liu, "The effects of integrating mobile devices with teaching and learning on students' learning performance: A meta-analysis and research synthesis," *Computers & Education*, vol. 94, pp. 252–275, 2016, doi: 10.1016/j.compedu.2015.11.008.
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- [12] K. Charan, "Class-Monitary-System," GitHub repository. [Online]. Available: <https://github.com/khushicharan2005/Class-Monitary-System>. Accessed: Sep. 25, 2026.

14. ANNEXURES AND SUBMISSION CHECKLIST

No.	Checklist Item	Yes / No	Page / Annexure
1	Synopsis follows the prescribed structure	YES	1-20
2	Problem statement is specific and evidence-backed	YES	5
3	Relevant literature has been reviewed and cited	YES	6-8, 17
4	Existing solutions have been compared	YES	9
5	Research / implementation gap is clearly established	YES	10
6	Stakeholder evidence is attached, where applicable	YES	11 / Annexure B
7	Scope, objectives and preliminary requirements are stated	YES	12-13
8	All tables, figures and external material are attributed	YES	4-17
9	References use a consistent citation style	YES	17
10	Similarity / originality report is attached, if required	YES	Annexure D

Suggested Annexures

Annexure A: Survey/questionnaire or interview schedule | Annexure B: Response/evidence summary | Annexure C: Supporting figures/screenshots | Annexure D: Similarity/originality report

15. STUDENT DECLARATION

I declare that this synopsis is my original academic work. All sources have been appropriately acknowledged. The facts and evidence presented have been verified to the best of my knowledge, and the submission complies with institutional academic-integrity requirements.

Student Name	Registration No.	Programme	Signature	Date
Ms. Khushi Charan	12522344	MCA		

16. FACULTY EVALUATION RUBRIC (30 MARKS)

Criterion	Excellent	Good	Developing	Needs Improvement	Max
Problem identification and significance	Precise, significant and strongly evidenced	Clear and relevant; adequate evidence	Broad or partially supported	Unclear or unsupported	6
Literature review and evidence	Relevant, recent and critically synthesized	Relevant sources with some synthesis	Limited sources; mostly descriptive	Insufficient or unreliable sources	6
Existing-solution analysis	Systematic, balanced comparison	Clear comparison with minor omissions	Basic listing; limited comparison	Missing or inaccurate analysis	5
Gap analysis and validation	Specific, logical and evidence-backed gap	Clear gap with adequate support	Gap stated but weakly supported	Gap absent or unjustified	5
Scope, objectives and requirements	Focused, feasible and well aligned	Mostly clear and aligned	Partial clarity or alignment	Vague or infeasible	5
Documentation, citations and presentation	Professional, coherent and accurate	Clear with minor issues	Several formatting/citation issues	Poorly organized or incomplete	3

Criterion	Marks Awarded
Problem identification and significance	/ 6
Literature review and evidence	/ 6
Existing-solution analysis	/ 5
Gap analysis and validation	/ 5
Scope, objectives and requirements	/ 5
Documentation, citations and presentation	/ 3
TOTAL	/ 30

17. FACULTY FEEDBACK AND APPROVAL FORM**Strengths of the Submission**

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Required Improvements / Corrective Actions

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Decision

Approved to Proceed to Task 2	☐ Yes ☐ Yes, after revisions ☐ Resubmit
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Faculty Mentor's Name	Mr. Manish Singh
Signature and Date	
Project / Course Coordinator	
Signature and Date	

Note: Students should preserve the evaluated synopsis and incorporate the approved findings into Academic Task 2: Proposed Solution and Prototype Design.